Data Science Notes

Workflow, data preparation, statistics, EDA, and visualization

1. Introduction and Workflow

Data Science combines programming, statistics, domain knowledge, and analytical thinking to extract insights from data and support decisions.

1. Define the problem and success criteria.
2. Collect data from files, databases, APIs, surveys, or sensors.
3. Clean and validate data.
4. Explore patterns with statistics and charts.
5. Build and evaluate an analysis or model.
6. Communicate findings and monitor results.

- Data analysis answers questions about data.
- Data engineering builds systems to collect, store, and move data.
- Machine learning learns patterns for prediction or decision support.
- Business intelligence presents metrics, reports, and dashboards.

2. Data Types, Sources, and Formats

- Structured data: tables with rows and columns.
- Semi-structured data: JSON or XML.
- Unstructured data: text, images, audio, and video.
- Numerical data: counts or measurements.
- Categorical data: labels such as department or product.

Common sources include CSV and Excel files, SQL databases, APIs, logs, surveys, IoT devices, and public datasets. CSV is tabular text; JSON supports nested key-value structures; SQL databases store related tables.

Check permissions, ownership, quality, and privacy before collecting or using data. Keep a raw copy before cleaning and document where data came from.

3. Data Cleaning and Preprocessing

Real-world data can contain missing values, duplicates, inconsistent labels, incorrect data types, and outliers. Investigate issues before changing records.

- Missing values: examine why values are absent; remove or impute only when justified.
- Duplicates: determine whether repeated records are genuine.
- Inconsistent categories: standardize spelling and capitalization.
- Incorrect types: parse dates and convert numeric columns correctly.
- Outliers: inspect context rather than automatically deleting them.
- Scaling: normalization or standardization may help some algorithms.

```
import pandas as pd
df = pd.read_csv("data.csv")
print(df.head())
print(df.isna().sum())
df = df.drop_duplicates()
print(df.describe(include="all"))
```

4. Exploratory Data Analysis and Statistics

Exploratory Data Analysis (EDA) uses summaries and visualizations to understand distributions, relationships, unusual observations, and data quality.

- Mean: sum divided by count; sensitive to extreme values.
- Median: middle value after sorting; often more robust to outliers.
- Mode: most frequent value.
- Range: maximum minus minimum.
- Variance: describes spread using squared deviations.
- Standard deviation: square root of variance.
- Correlation describes association, not necessarily causation.
- Univariate analysis studies one variable.
- Bivariate analysis studies a relationship between two variables.
- Multivariate analysis examines several variables together.

5. Data Visualization

- Bar chart: compare categories.
- Histogram: show a numerical distribution.
- Line chart: show change over time or ordered values.
- Scatter plot: show the relationship between two numerical variables.
- Box plot: compare distributions and flag potential outliers.
- Heatmap: display values in a matrix, often correlations.

Use a clear title, labelled axes, units, readable scales, and accessible colours. Avoid misleading scales and unnecessary decoration.

```
import matplotlib.pyplot as plt
plt.hist(df["age"].dropna(), bins=10)
plt.title("Age Distribution")
plt.xlabel("Age")
plt.ylabel("Count")
plt.show()
```

6. Machine Learning, Ethics, and Mini Projects

- Regression predicts numerical values; classification predicts categories; clustering groups similar observations.
- Split data into training and test sets. Fit preprocessing steps on training data only to avoid data leakage.
- Protect personal information and use data only with appropriate permission.
- Check assumptions, bias, uncertainty, and limitations before communicating conclusions.
- Mini-project ideas: student marks analysis, monthly sales trends, weather visualization, customer feedback exploration, or traffic pattern analysis.